

Alpha Return Dynamics: Evidence from Hedge Fund Indices

Gregory Koutmos

Discussant: Boone Bowles

February 28, 2024

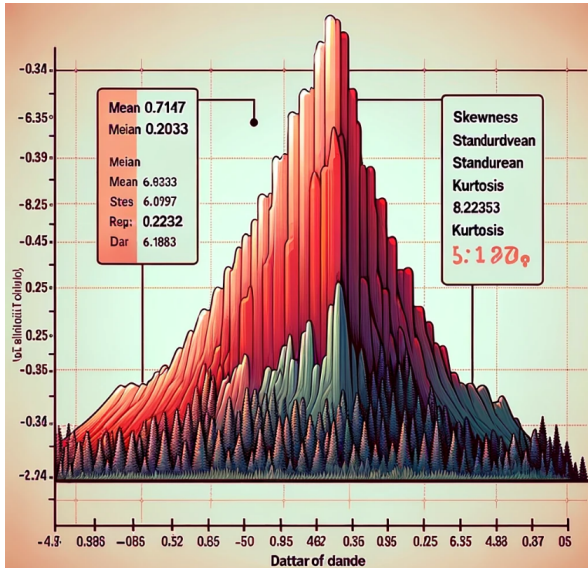
SWFA 2024

The paper in one slide.

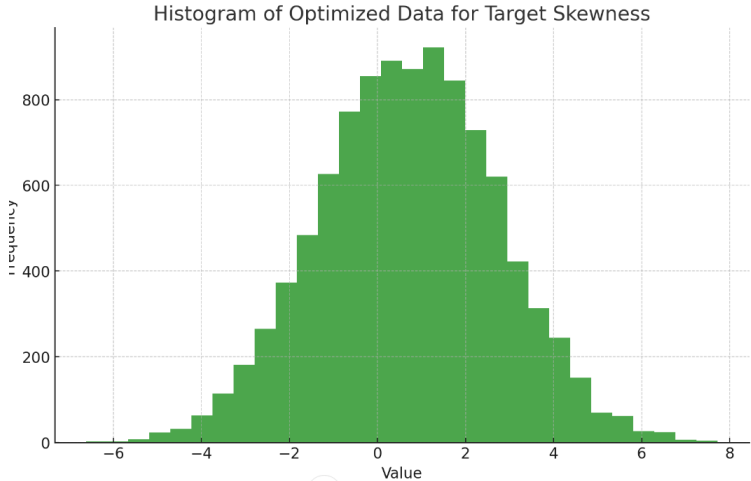
- ① Hedge fund managers get paid for generating alpha, so how robust is alpha?
- ② Hedge Fund alphas **are not stable** across the distribution of returns.
 - When HFs do really well or really bad it's not alpha.
 - When HFs do average (between -1% and $+2\%$ per month) some of that is alpha.

Accessible paper, clear evidence, highly recommended! It made me really interested in this datasource.

The paper in three pictures.

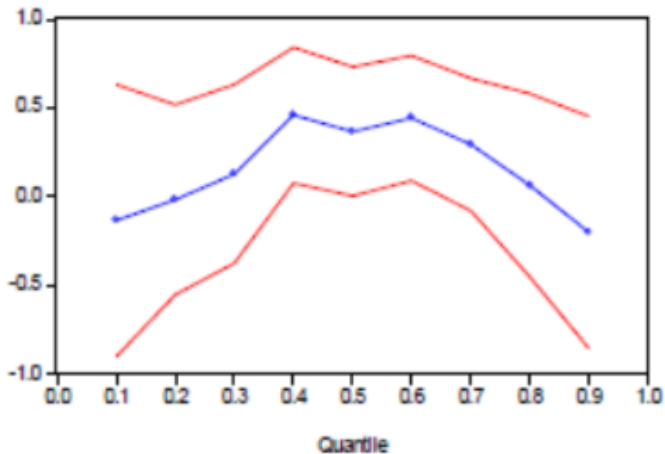


The paper in three pictures.



The paper in three pictures.

alpha



Major Point 1

What is going on in the extreme times?

Major Point 1

What is going on in the extreme times?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

Major Point 1

What is going on in the extreme times?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (\alpha/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

- Is volatility higher?
- Did MOM have a big month?
- If it's not α then what is it?

Major Point 2

What is going on in the normal times?

Major Point 2

What is going on in the normal times?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

Major Point 2

What is going on in the normal times?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
a	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
b_m	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
b_{SMB}	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
b_{HML}	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
b_{MOM}	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

- Is the market more calm?
- Is it outside or inside of earnings season?
- What is it about these months that allows alpha?

Major Point 3

How are the extreme months different from the average months?

Major Point 3

How are the extreme months different from the average months?

- Why is MOM a big deal in the extreme months?
- Good times seem to be correlated with SMB too?

Two Less Major Points

1 - Can you show the standard errors?

Two Less Major Points

1 - Can you show the standard errors?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

Two Less Major Points

1 - Can you show the standard errors?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

- The SEs get larger in the extreme months, what does that mean for alpha?
- Are monthly returns too frequent to measure skill?

Two Less Major Points

2 - Can you produced Table 5 for the various subcategories of HF strategies?

Two Less Major Points

2 - Can you produced Table 5 for the various subcategories of HF strategies?

Table 5. Quantile Regression with Scaled Variables (for Hedge Fund Index HFI)

$$(r_{i,t}/\sigma_t) = (a/\sigma_t) + \varphi(r_{i,t-1}/\sigma_t) + b_m(r_{m,t}/\sigma_t) + b_{SMB}(SMB_t/\sigma_t) + b_{HML}(HML_t/\sigma_t) + b_{MOM}(MOM_t/\sigma_t) + (\varepsilon_t/\sigma_t)$$

quantile	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
<i>a</i>	-0.1374	-0.0205	0.1258	0.4571	0.3654	0.4416	0.2915	0.0622	-0.2037
t-stat	-0.350	-0.074	0.488	2.341*	1.963*	2.451*	1.528	0.235	-0.608
<i>b_m</i>	0.3339	0.2842	0.2949	0.2825	0.2663	0.2646	0.2492	0.2539	0.2559
t-stat	5.342*	9.705*	10.587*	11.030*	11.073*	12.265*	12.165*	9.116*	9.478*
<i>b_{SMB}</i>	0.0174	-0.0002	-0.0139	-0.0020	0.0223	0.0196	0.0398	0.0306	0.0357
t-stat	0.303	-0.005	-0.358	-0.055	0.755	0.692	1.433	0.896	0.954
<i>b_{HML}</i>	0.0667	0.0271	0.0598	0.0455	0.0111	0.0255	0.0164	0.0124	-0.0212
t-stat	1.221	0.585	1.269	1.049	0.317	0.861	0.599	0.387	-0.704
<i>b_{MOM}</i>	0.1819	0.1402	0.1268	0.0871	0.0798	0.0733	0.0808	0.0945	0.1190
t-stat	3.678*	4.040*	3.209*	2.528*	3.537*	4.109*	5.279*	5.942*	9.059*

- I'm really interested to see if alpha is more robust for market neutral or long/short equity.

Recap in one slide.

Very nice paper. Clear findings. Taught me something.
Recommended reading.

TONS of potential to teach us some more cool things!